

PROBLEM STATEMENT #34 | Retail, E-Commerce & Finance

Digital Gift Card & Loyalty Point Exchange

1. Problem Context & Overview

A digital wallet platform allowing users to aggregate brand loyalty points, convert points into universal exchange credits at dynamic rates, and redeem digital gift cards with anti-fraud voucher locking.

Target Stakeholders / Actors: *Account Holder, Merchant Partner*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.

Sample Acceptance Criteria: Pass: Voucher code generated with cryptographic checksum; Fail: Expired or duplicate voucher redeemed.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Point conversion and gift card redemption operations must process within 250 ms with zero ledger balance drift.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.